

Zero-Dimensional Internal Ballistics

The objective of this coursework is to apply the zero-dimensional internal ballistic model to VEGA's P80 solid rocket motor, which external case and internal grain geometry (finocyl) is shown in fig.1. One starts from the simplest analysis and then the complexity of the evaluation of the chamber pressure is increased by adding the role of temperature variation and non-ideal parameters.

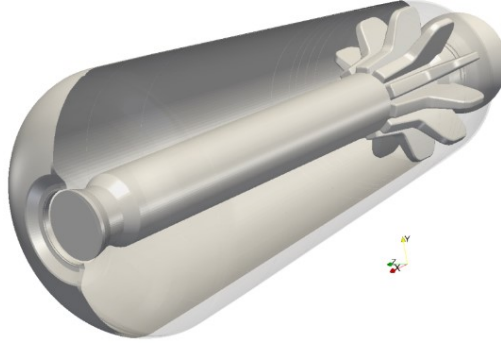


Figure 1: P80 solid rocket motor finocyl geometry.

Step 1: Quasi-steady state

The analysis is taken considering a negligible erosive burning contribution to the burning rate and the validity of St. Robert's burning rate law:

$$r = a p_c^n \quad \text{where} \quad a = a_0 \exp[\tau(T_i - T_{i0})] \quad (1)$$

where they appear, in addition to the chamber pressure p_c and the burning rate r : the temperature coefficient a , the combustion index n , the temperature sensitivity τ and the reference temperature T_{i0} .

In order to evaluate the evolution of chamber pressure, more assumptions are required in addition of the general one for the zero-dimensional model:

1. combustion products are assumed as an ideal gas with constant thermodynamic properties (i.e. thermally and calorically perfect gas);
2. chamber conditions are assumed to be uniform in terms of p_c , T_c and ρ_c ;
3. nozzle throat is assumed to be choked.

So, using all the previous assumptions, from the mass balance equation inside the motor chamber volume V_c one can obtain the differential equation:

$$\frac{dm_c}{dt} = \frac{d(\rho_c V_c)}{dt} = \dot{m}_t - \dot{m}_b = \rho_p S_b a p_c^n - \frac{p_c A_t}{c^*} \quad (2)$$

that can be rearranged assuming that $dV_c/dt = r S_b$ and obtaining:

$$V_c \frac{d\rho_c}{dt} = (\rho_p - \rho_c) S_b a p_c^n - \frac{p_c A_t}{c^*}$$

At the end, the ODE for p_c is obtained assuming, for this first step, that the chamber pressure T_c has no influence on p_c (i.e. $dT_c/dt = 0$):

$$\frac{dp_c}{dt} = \frac{p_c}{\tau} \left[\left(\frac{p_{ceq}}{p_c} \right)^{1-n} - 1 \right] \quad (3)$$

paying attention that the τ that appears here is not the same as eq.1, this is the characteristic time scale of the motor, defines as:

$$\tau = \frac{1}{\Gamma^2 c^*} \frac{V_c}{A_t} \quad (4)$$

and it describes the dynamic response of the motor because the ratio between V_c and A_t is changing in time.

The equilibrium chamber pressure is obtained putting $dp_c/dt = 0$ and assuming $\rho_c \ll \rho_p$ in eq.2.

$$\dot{m}_t = \dot{m}_b \longrightarrow p_{ceq} = (\rho_p a c^* K)^{\frac{1}{1-n}} \quad (5)$$

where $K = S_b/A_t$ is a fundamental geometrical parameter called Klemmung ratio.

In general, for the zero dimensional model is more convenient to describe the pressure variation as a function of the burned web thickness rather than of time. This is due to the fact that the burning time of a SRM can change easily depending on the operational conditions, while the burned web remains unchanged for its exclusive dependency on the grain geometry. In the analyzed case, in fact, the burning surface is given as a function of the burned web thickness and it is shown in fig.2.

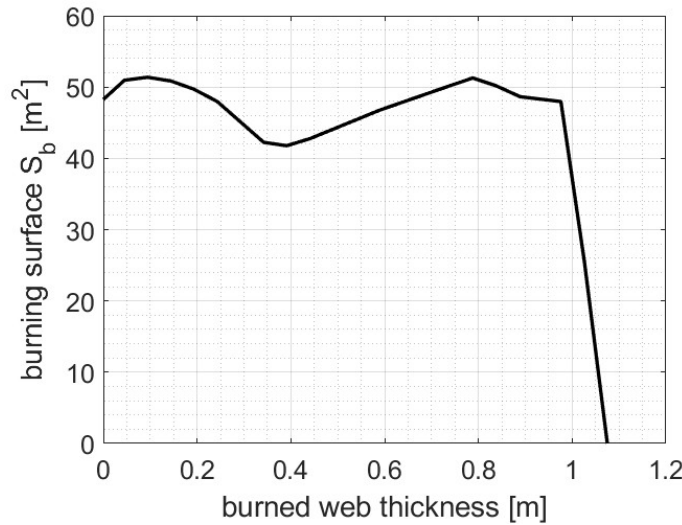


Figure 2: P80 burning surface evolution vs web thickness.

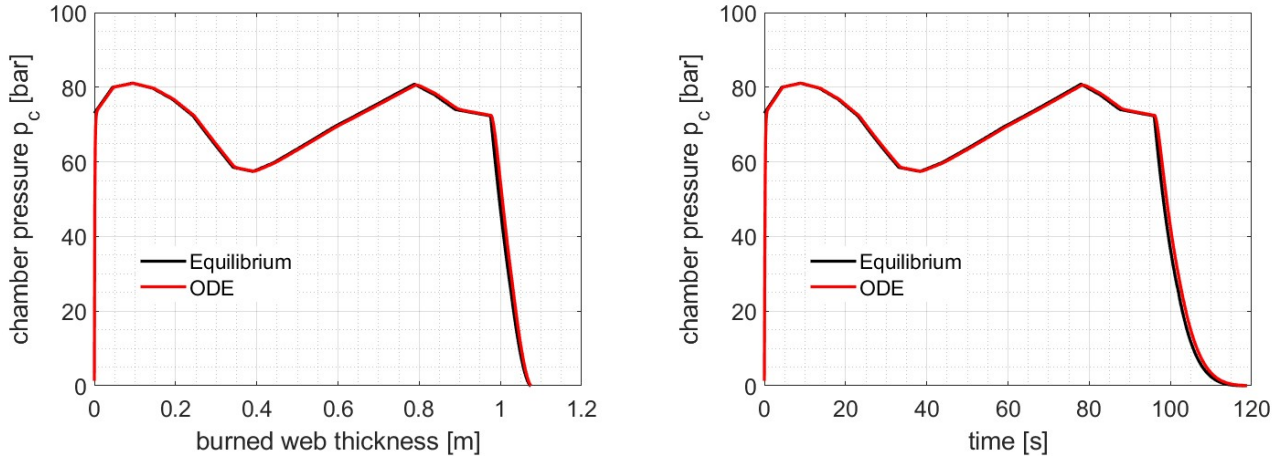
But, in order to evaluate the requested variables as a function of time, one has to relate it with the given web thickness following the expression:

$$\frac{dw}{dt} = r \longrightarrow t = \int_0^w \frac{1}{ap_c^n} dW \quad (6)$$

Thanks to this relation, one can evaluate the evolution of the chamber pressure as a function of the web, integrating the expression:

$$\frac{dp_c}{dw} = \frac{1}{\tau a} [(\rho_p - \rho_c) a c^* K - p_c^{1-n}] \quad (7)$$

Finally, the results can be evaluated and they are shown in fig.3. The web evolution of the equilibrium chamber pressure can be computed using the expression 5, which is directly a function of the web because of the dependency of the burning surface from it, while the time evolution is easily obtained plotting it as a function of the integrated time from eq.6. The chamber pressure evolution computed from the solution of the ODE refers to time differential equation 3 and web differential eq.7.



Chamber pressure as a function of web thickness.

Chamber pressure as a function of time.

Figure 3: P80 pressure profiles.

Once the chamber pressure is evaluated, the vacuum thrust evolution is obtained following the quasi-one-dimensional theory of the ideal nozzle:

1. Evaluation of the expansion ratio from the given area ratio $\epsilon=16$ solving the non linear equation taken from the expression:

$$\epsilon = \frac{A_e}{A_t} = \frac{\Gamma}{\sqrt{\frac{2\gamma}{\gamma-1} \left(\frac{p_e}{p_c}\right)^{\frac{2}{\gamma}} \left[1 - \left(\frac{p_e}{p_c}\right)^{\frac{\gamma-1}{\gamma}}\right]}} \longrightarrow \frac{p_e}{p_c} = 0.085$$

2. Substituting the expansion ratio in the expression of the vacuum thrust coefficient:

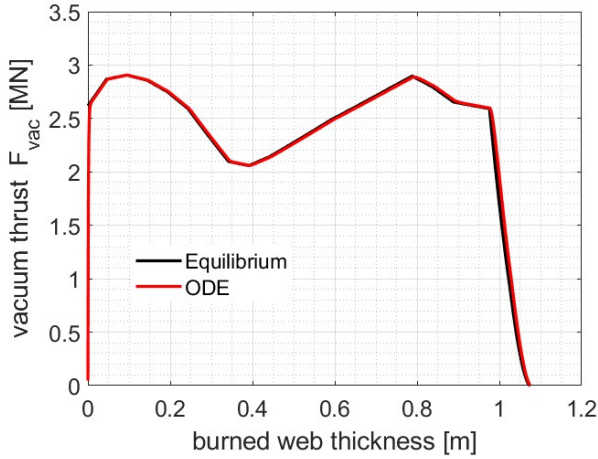
$$C_{F_{vac}} = \Gamma \sqrt{\frac{2\gamma}{\gamma-1} \left[1 - \left(\frac{p_e}{p_c} \right)^{\frac{2}{\gamma}} \right]} + \epsilon \left(\frac{p_e}{p_c} \right) = 1.855$$

3. Compute the vacuum thrust as:

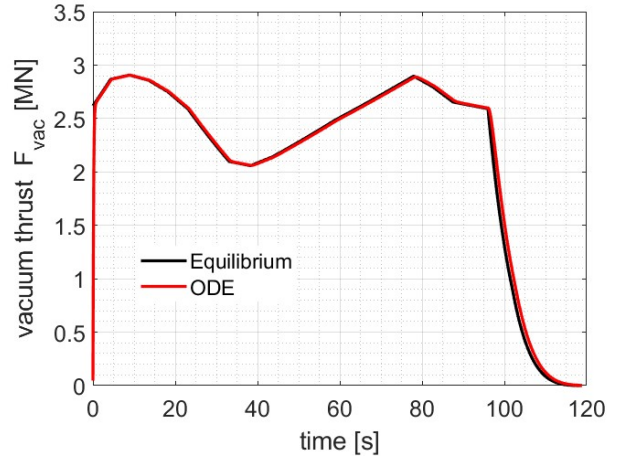
$$F_{vac} = p_c C_{F_{vac}} A_t$$

where the dependency of the obtained expression is driven by the choice of the expression of p_c , as the equilibrium one or the one computed from the solution of the ODE equation as a function of web or time.

The results are shown in fig.4.



Vacuum thrust as a function of web thickness



Vacuum thrust as a function of time

Figure 4: P80 vacuum thrust profiles.

Since this first step is based on the isothermal model, i.e. T_c does not vary, and the same is valid for the characteristic velocity c^* , the specific impulse remains the same during all the operational time and the vacuum one can be computed as: $I_{sp_{vac}} = (C_F c^*)/g_0 = 300.75$ s, which is significantly higher than the one indicated in the given table of all the motor's parameters because of its non-ideal behaviour.

In order to evaluate the sea level performances, one needs the sea level thrust coefficient, which can be easily computed from the vacuum one as:

$$C_{F_{sl}} = C_{F_{vac}} - \epsilon \frac{p_{a_{sl}}}{p_c}$$

and using a reference pressure given by the value of the equilibrium pressure at $t=0$, $p_{cref} = 73 \text{ bar}$, the results are:

$$F_{sl} = 2.3 \text{ MN} \quad I_{sp_{sl}} = 264.78 \text{ s}$$

For what concerns the variation of the gases in the chamber over time, starting from the left part of eq.2, one can separate the role of density variation to the one of chamber volume:

$$\frac{dm_c}{dt} = \frac{d}{dt}(\rho_c V_c) = V_c \frac{d\rho_c}{dt} + \rho_c \frac{dV_c}{dt} \quad (8)$$

$$\text{where : } \rho_c \frac{dV_c}{dt} = \rho_c S_b(t) a p_c^n \quad \text{and} \quad V_c \frac{d\rho_c}{dt} = (\rho_p - \rho_c) S_b a p_c^n - \frac{p_c A_t}{c^*} \quad (9)$$

the resultant plot is shown in figure 5:

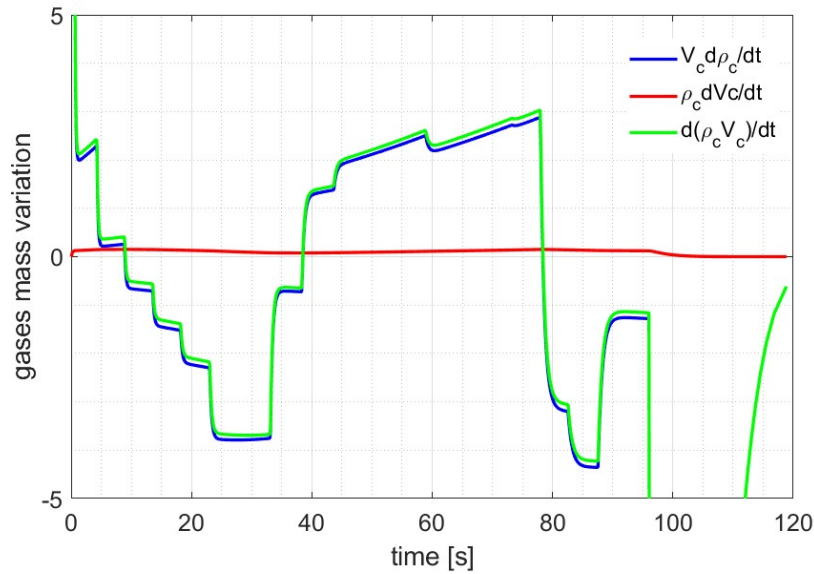


Figure 5: P80 gases mass variation

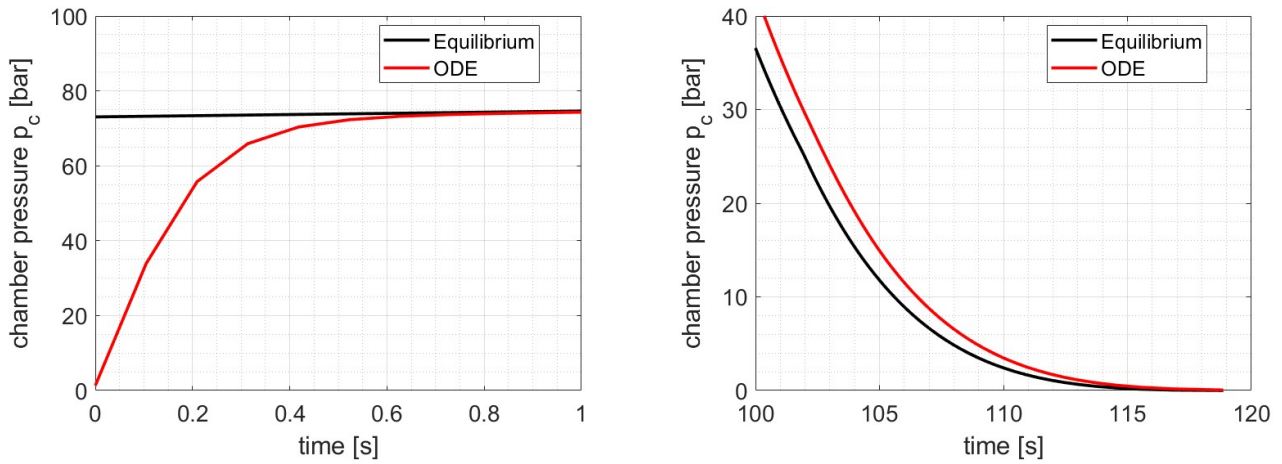
Comments and explanation of the results

The pressure profiles and thrust profiles plots show a very similar behaviour as the burning surface evolution, which is predictable since it is the only parameter that is varying either with burned web thickness case or the one that refers to the time.

The figures highlight the overlap between the model which relates to the equilibrium pressure and the one related to the solution of the ODE equation. Obviously, the similarity does not appear for all the abscissa interval, either web or time, but it is evident mostly in the regions where $dp_c/dt \simeq 0$ (i.e. quasi-steady state condition).

In particular, looking to the time intervals where the pressure variation is higher, the so called transient phases, showed in figure 6, one can conclude that the integrated and equilibrium pressure chase each other with a trend driven by the sign of dp_c/dt :

- when $\frac{dp_c}{dt} < 0$ the equilibrium chamber pressure is lower than the integrated (i.e. the ODE p_c follows the p_{ceq});
- when $\frac{dp_c}{dt} = 0$ the equilibrium chamber pressure is equal to the integrated one;
- when $\frac{dp_c}{dt} > 0$ the equilibrium chamber pressure is higher than the integrated one (i.e. the p_{ceq} follows the ODE p_c).



Zoom on the chamber filling phase of the pressure vs time evolution.

Zoom on the tail-off phase of the pressure vs time evolution.

Figure 6: Zoom of the transient phases of P80 pressure profile.

Moreover, it can be noticed that the pressure build-up is faster than the pressure decrease during the tail-off. This is due to the role of the characteristic time scale τ which has been defined in eq.4. It is smaller at the ignition because of the smaller chamber volume, which corresponds to a fast-responding motor, against a slow-responding motor at tail-off.

It is obvious that all these considerations are valid also for the thrust variation, since it follows the behaviour of p_c , which is the only varying factor.

From fig.5 one can evaluate the role of density and chamber volume variations on gases mass variation. In particular the variation of the gases mass is driven only by the density variation and this is due to the fact that the density of the combustion gases ρ_c is very small with respect to that of the propellant ρ_p and this reflects on the fact that the term $\rho_c \frac{dV_c}{dt}$ in eq.8 becomes negligible compared to the other.

The effect of initial grain temperature T_i is shown in fig.7, where the chamber pressure is evaluated

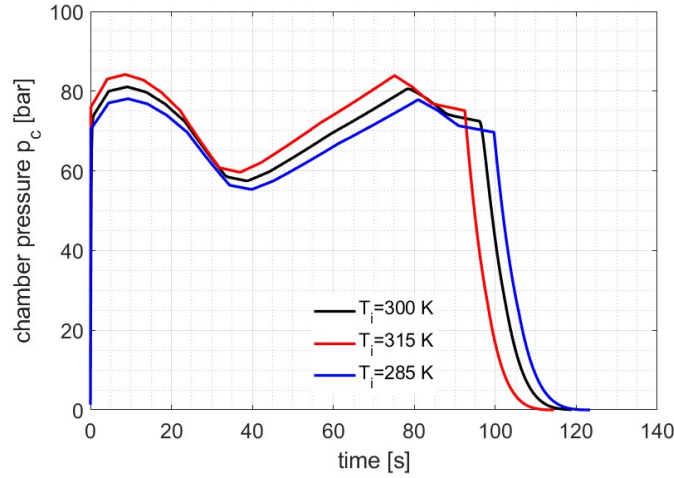


Figure 7: Influence of the grain temperature T_i on the chamber pressure as a function of time.

considering, in addition to the standard case where $T_i = 300K$, a variation of $\pm 15K$. The two main effects are:

- an increase in T_i shows an higher chamber pressure during most of the operational time and this is due to the dependency of the temperature coefficient a on the initial propellant temperature, which results to an increasing value of p_c from the St. Robert's burning rate law;
- the previous effect also has as a consequence a smaller burning time because the propellant is burning faster.

Step 2: Energy equation

The goal of this step is to neglect the previous heavy assumption of isothermal model and re-evaluate the variation of the chamber pressure and vacuum thrust either in the web and time. Now T_c is allowed to change, even though the chamber pressure variation effects on thermochemical equilibrium calculation are still neglected.

So, one must introduce the energy balance equation in addition to the mass balance equation considered up to now. By neglecting the kinetic energy contribution to the total energy, one obtains the following two ODE equations in the p_c and T_c variables:

$$\begin{cases} \frac{dp_c}{dt} = \frac{\gamma R}{V_c} (\dot{m}_b T_{af} - \dot{m}_c T_c) - \frac{p_c}{V_c} \frac{dV_c}{dt} \\ \frac{dT_c}{dt} = \frac{RT_c}{p_c V_c} [\gamma (\dot{m}_b T_{af} - \dot{m}_t T_c) - (\dot{m}_b - \dot{m}_t) T_c] \end{cases}$$

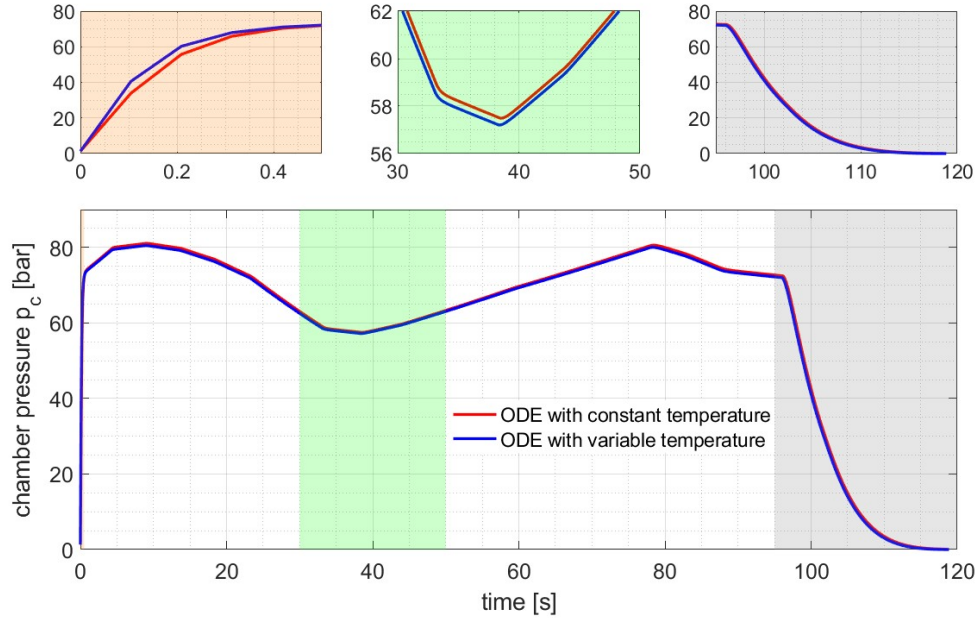


Figure 8: P80 chamber pressure evolution over time for different models with zooms in the region of ignition transient, maximum dynamic pressure and tail-off phases.

The chamber pressure integration result is shown in fig.8, where it is compared with the previous isothermal model.

One can underline the fact that, especially in the chamber filling phase, highlighted in light orange in the figure, the derivative of T_c plays an important role: in the early phase of the life of the motor, where both p_c and T_c are increasing quickly in time, the rise in time of the chamber pressure is increased by the contribution of the chamber temperature evolution. This fact is clearly visible by the evolution in time of T_c , showed in fig.9, where at ignition there is a fast increasing phase with an overshooting zone, followed by the middle zone where the value of T_c does not exceed much the adiabatic flame temperature and at the end, the depressurization phase, where the temperature decreases as well as pressure.

The vacuum thrust follows the same trend of the chamber pressure, as one can see in fig.10.

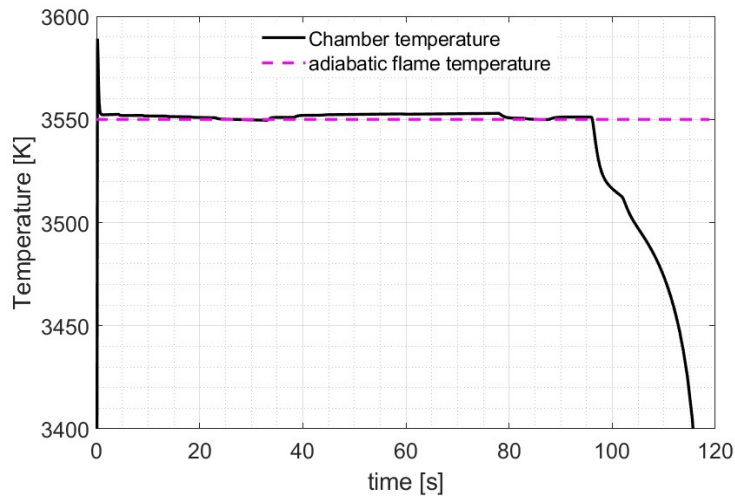


Figure 9: P80 chamber temperature evolution over time for the non-isothermal model

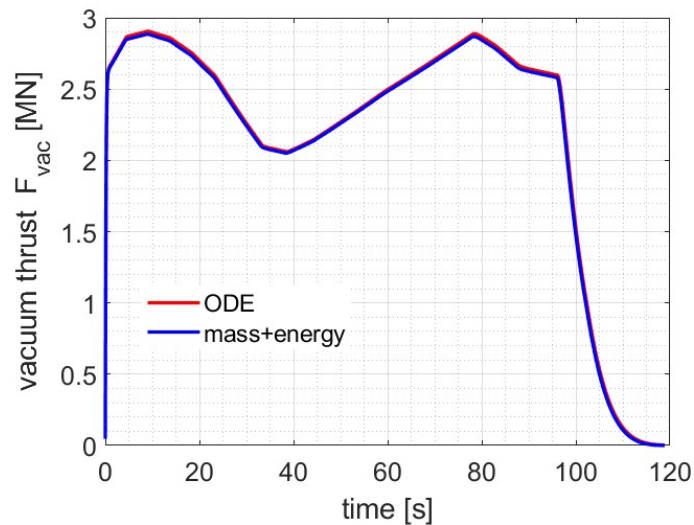


Figure 10: P80 vacuum thrust evolution over time for different models

Step 3: Non-ideal parameters

From the previous models, one can reconstruct the pressure-time curve in a solid rocket motor, but the real behaviour is affected by some non-ideal phenomena which are enclosed in the so called *non-ideal parameters* and they help to correct the zero-dimensional model in order to give results which are as close to reality as possible.

They are:

- combustion efficiency or c^* efficiency, η_{c^*} ;

- thrust efficiency or C_F efficiency, η_{C_F} ;
- scale factor SF ;
- *HUMP* effect;
- throat erosion.

Combustion efficiency

This parameter represents a correction for the combustion process, which in the real case is different from what it is evaluated with the ideal value c^* , where complete and adiabatic combustion is considered. In particular the real c^* will be lower than the ideal one:

$$\eta_{c^*} = \frac{c^*}{c_{id}^*} < 1 \quad \text{where} \quad \eta_{c^*} = \frac{\int_0^{tb} \frac{p_c(t)A_t(t)}{c_{id}^*} dt}{m_p}$$

and this comes from the ratio between the integration of mass flow rate given by the ideal expression, which comes from the ideal value of c^* , and the real propellant mass, which are clearly different and the combustion efficiency helps to get closer the ideal value to the real one.

Thrust efficiency

The thrust efficiency is the only parameter that does not affect the chamber pressure but only the thrust evolution, since C_F is related to the expansion process given by the nozzle design.

It is represented by the ratio between the instantaneous thrust coefficient evaluated from the thrust expression and the ideal C_F .

$$\eta_{C_F} = \frac{\frac{F(t)}{p_c(t)A_t(t)}}{C_{F_{id}}}$$

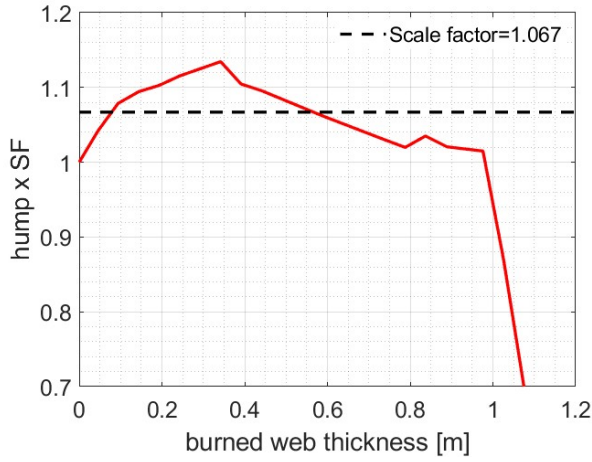
Scale factor and HUMP effect

They are related to the propellant burning rate, in particular with the temperature coefficient a . Their introduction is needed to take into account the mismatch between the propellant characterization operated through strand burner test and the full-scale motor burning rate, resulting into:

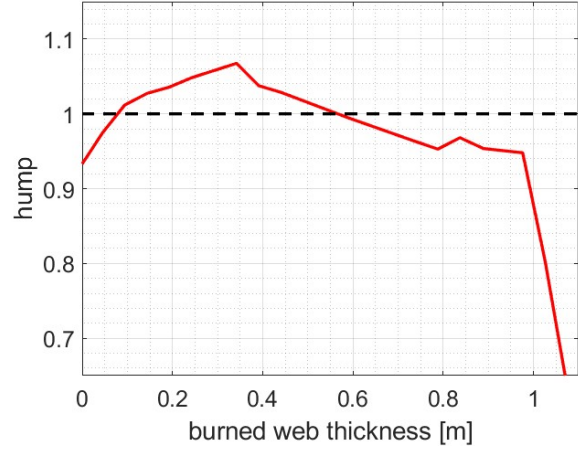
$$r_{motor} = HUMP(web) \times SF \ a \ p_c^n(web)$$

Typically, the HUMP and the scale factor are extracted from data as a single function of the web thickness: to separate the two effects, the average value of the function $HUMP \times SF$ over web is calculated (which is greater than 1) and is taken the SF of the motor, while the rest is attributed to the HUMP effect only (which has an average value equal to 1).

In this case the evolution of the function $HUMP \times SF$ is given and the previous operations are performed in fig.11.



P80 HUMP $\times$ SF evolution vs web thickness.



P80 HUMP evolution vs web thickness.

Figure 11: Extrapolation of the HUMP from the HUMP $\times$ SF evolution.

Throat erosion

The throat erosion effect implies that the throat diameter is not constant over time, but increases because the throat region is protected by ablative materials. Consequently, a correction in A_t inside the equilibrium chamber pressure expression must be taken into account. In particular it is assumed a linear regression in time of throat radius:

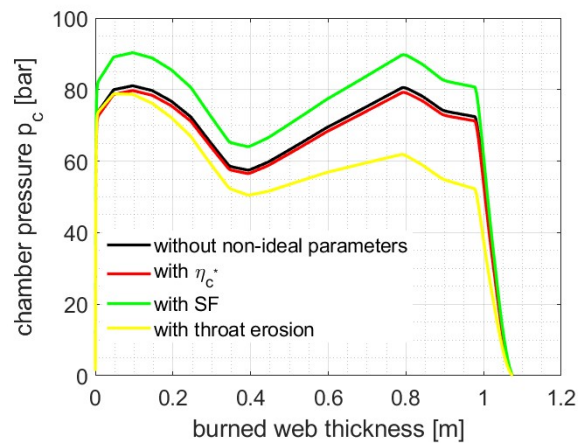
$$D_t(t) = D_{t_{in}} + 2\dot{s}t \longrightarrow A_t(t) = \pi \frac{D_t^2(t)}{4}$$

where $\dot{s}$ is the given erosion rate.

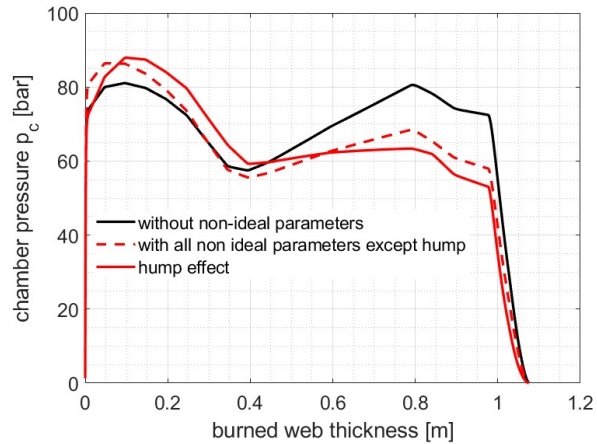
The effect of the introduction of each non-ideal parameter, one by one, is now investigated.

In fig.12, fig.13 and fig.14 there are the results. The reference model indicated with *without non-ideal parameter* is the isothermal one. All the following explanations should be considered valid both for the chamber pressure variation and vacuum thrust variations, but those referring to the burning time variations must refer to the plots with time as the abscissa.

- The c^* **efficiency** plays a role of a scaling factor of the overall pressure behaviour and the red curve can be obtained as a simple shift from the black one. Moreover, a delay in the burning time is added by η_{c^*} as a consequence of the decrease of p_c , and thus of a smaller burning rate.
- The **scale factor** taken alone, has only a translation effect of the reference chamber pressure of the 6.7%, which is its value, clearly explained in fig.11. This results in a decrease in burning time.
- The **throat erosion** effect introduced a progressive decrease in the pressure values over time. This is due to the continuous decrement of the Klemmung ratio which follows an

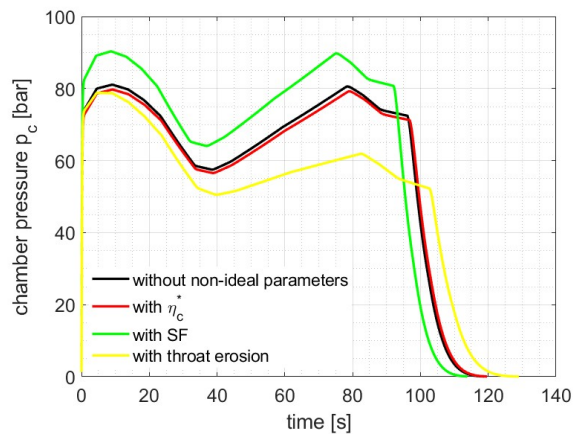


Chamber pressure as a function of web thickness for all parameters excepts hump.

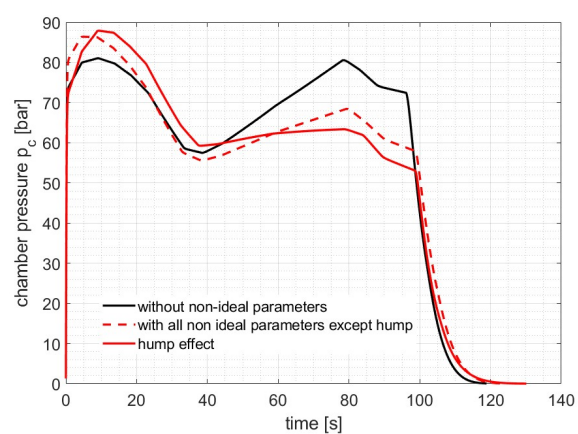


HUMP effect on the chamber pressure as a function of web thickness.

Figure 12: P80 pressure profiles vs web (including non-ideal parameters).

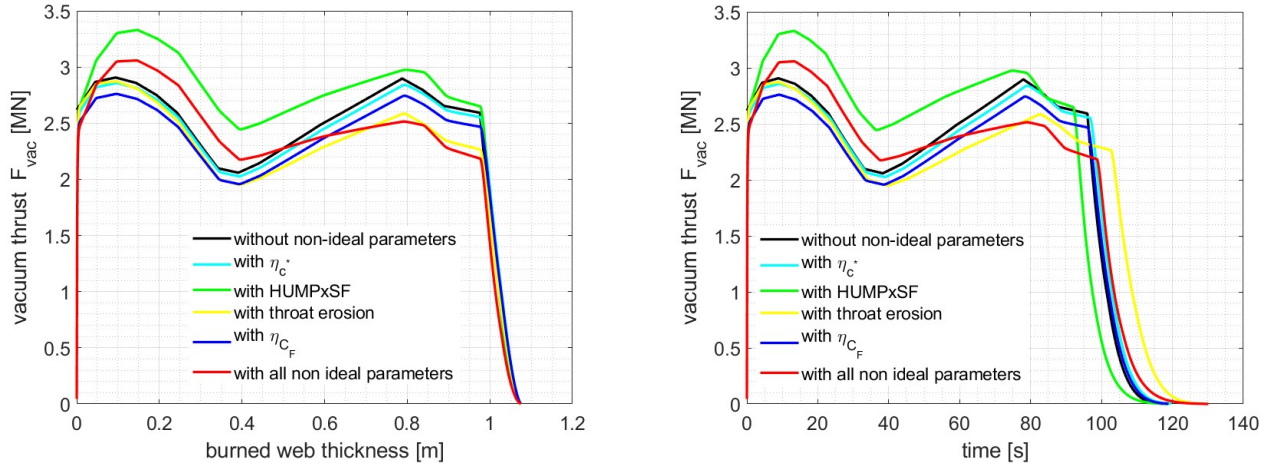


Chamber pressure as a function of time for all parameters excepts hump.



HUMP effect on the chamber pressure as a function of time.

Figure 13: P80 pressure profiles vs time (including non-ideal parameters).



Vacuum thrust as a function of web thickness. Vacuum thrust as a function of web thickness.

Figure 14: P80 vacuum thrust profiles (including non-ideal parameters).

increase of A_t . This translates in a smaller burning rate and thus in a longer-burning motor. Moreover, in this case, the vacuum thrust curve has a different behaviour with respect to the chamber pressure one. This depends on the linear dependence on A_t of F_{vac} and the thrust coefficient variation. The last one comes from the updating expansion ratio, which determines a decreasing $C_{F_{vac}}$. Nevertheless the linear A_t contribution prevails and the overall thrust values are not so far from the other models.

- The **HUMP** is the last corrective term, which is applied once all the other non-ideal parameters are taken into account. In this case one has to look to the right picture of fig.12 and fig.13. When all the parameters are taken into account except for the HUMP, the resulting curve is very similar to the experimental data, except for some points where the curve is higher or lower. The role of the HUMP factor is to adjust this trend and "rotate" the curve in such a way to best overlap the real behaviour. In particular it introduces an overpressure in the central portion of operations and an underpressure in the initial and final parts.
- The C_F **efficiency** is visible only in the evolution of thrust, as clearly explained before. It represents only a scaling factor with respect to the base model.

Motor specific impulse

The motor vacuum specific impulse can be evaluated using the following expression, taking into account the non-ideal parameters:

$$I_{sp_{vac}} = \frac{\eta_{C_F} C_{F_{vac}} \eta_{c^*} c^*}{g_0} = 282.86 \text{ s}$$

This is an instantaneous value, which does not take into account the time variation of all the non-ideal terms.

The best way to compute it is to evaluate its variation in time, taking into account all the contributions of the non-ideal parameters, doing an average between the initial value and final value at burnout. The resultant curves are shown in fig.15, where the constant value without non ideal parameters coincides with the green curve. From the final red curve, one gets an average value of

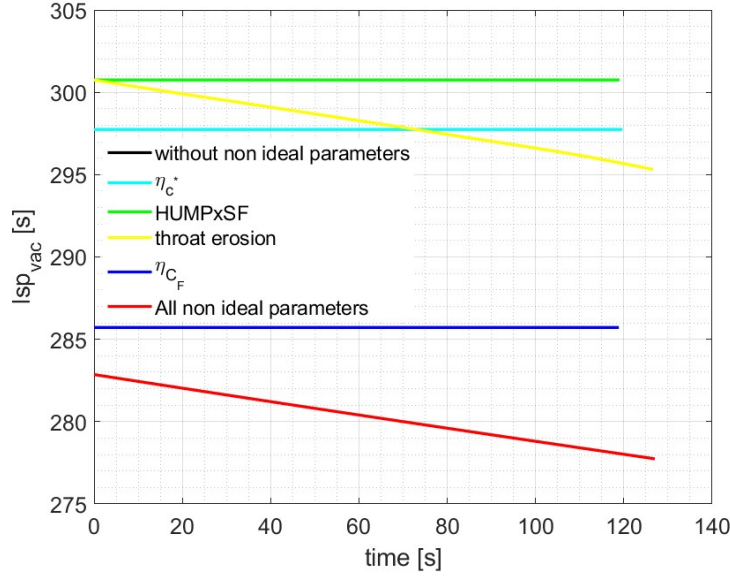


Figure 15: P80 vacuum specific impulse variation in time (including non ideal parameters).

vacuum specific impulse of $I_{sp_{vac}} = 280.3 \text{ s}$, which is very similar to the given real value.

For what concerns the variation of the specific impulse related to each non-ideal parameter, one can conclude that the only parameter that really change the trend is the throat erosion, which leads to a linear decreasing curve, due to the decrease of the thrust coefficient, and a greater burning time. All the others non-ideal parameters do not influence the time behaviour of the specific impulse but only the starting overall value.

The thrust coefficient C_F behaviour can be evaluated as a function of area ratio, introducing the throat erosion by limiting the evaluation from the initial area ratio to the final one. So, plotting the C_F value, in a linear scale, as a function of ϵ , in a logarithmic scale, for different values of the ratio p_c/p_a , where p_c is the chamber pressure and p_a is the ambient pressure, one can evaluate both the behaviour of the sea level and vacuum thrust coefficient. The result is shown in fig.16.

This leads to a decreasing behaviour of the $C_{F_{vac}}$, represented by the blue curve, which is the one with p_c over p_a equal to infinity, as $p_a = 0$. The sea level thrust coefficient is showed by the green curve, which is also a decreasing curve from the initial area ratio of 16 to the final one of 12.9, with a different trend of the vacuum one because it depends also on the variation of p_c and the fixed value of the ambient pressure. The red curves represents the different trends of C_F with an increasing value of the ratio between p_c and p_a from left to right. They are stopped when the nozzle reaches the separation.

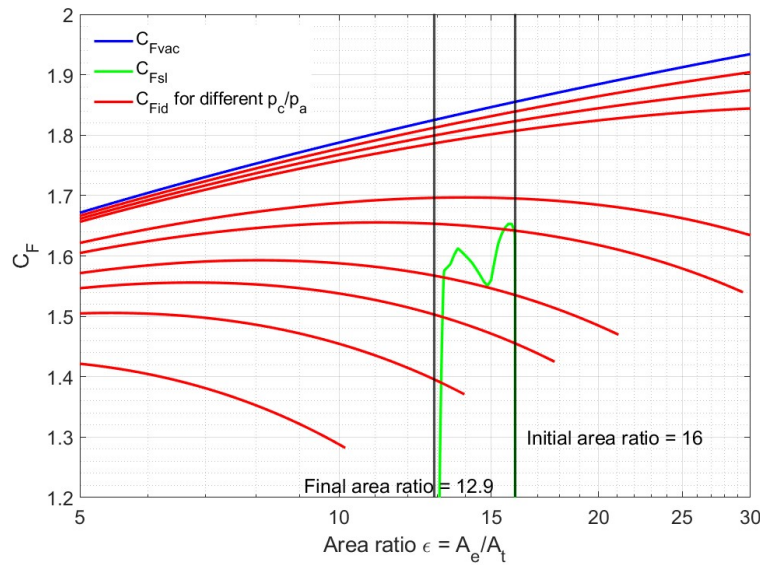
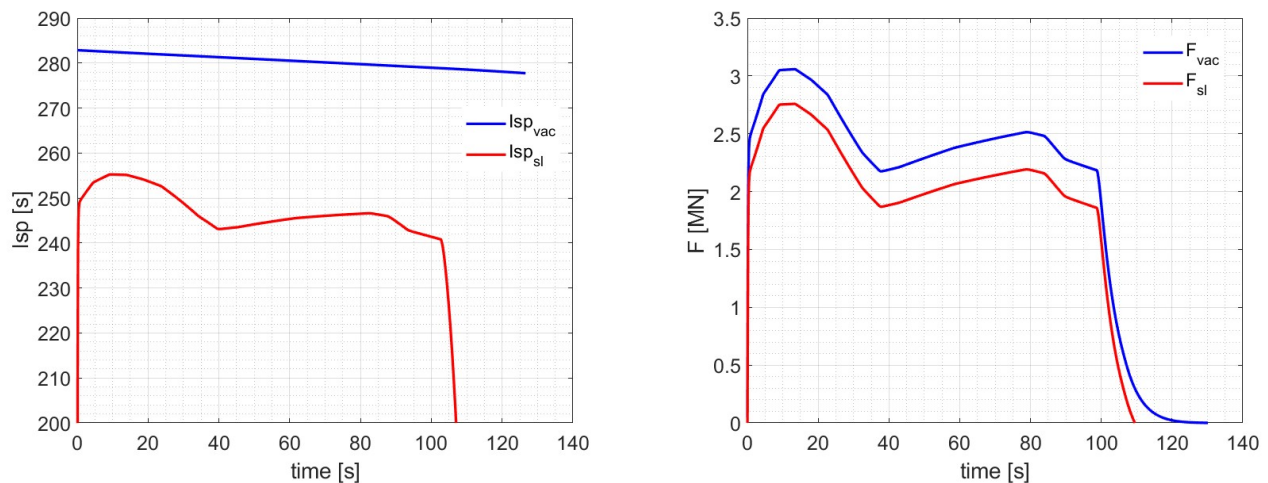


Figure 16: Thrust coefficient variation as a function of area ratio for different ratios between chamber pressure and ambient pressure.

Once the C_F is clearly evaluated, one can express also the sea level specific impulse as a time function, as well as the sea level thrust. They are shown in fig.17.



P80 vacuum and sea level specific impulse as a function of time.

P80 vacuum and sea level thrust as a function of time.

Figure 17: P80 vacuum and sea level performances.